Pre-trained Language Models Learn Remarkably Accurate Representations of Numbers

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Abstract

Pretrained language models (LMs) are prone to arithmetic errors. Existing work showed limited success in probing numeric values from models' representations, indicating that these errors can be attributed to the inherent unreliability of distributionally learned embeddings in representing exact quantities. However, we observe that previous probing methods are inadequate for the emergent structure of learned number embeddings with sinusoidal patterns.

In response, we propose a novel probing technique that decodes numeric values from input embeddings with *near-perfect accuracy* across a range of open-source LMs. This proves that after the sole pre-training, LMs represent numbers with remarkable precision. Finally, we find that the embeddings' precision, judged by our probe's accuracy, explains a large portion of LM's errors in elementary arithmetic, and show that aligning the embeddings with the pattern our probes discover can mitigate these errors.

1 Introduction

The landmark paper of Brown et al. (2020) showed that generic neural networks trained on text prediction alone could develop surprising arithmetic capabilities. In the years since, this observation has flourished into a large and vibrant field interested in the arithmetic reasoning capabilities of Transformers (Ahn et al., 2024), rife with research opportunities ranging from interpretability work (Akter et al., 2024) to solving Olympiad-level problems in mathematics (Li et al., 2025). Yet this work has also underscored the limitations of LMs on arithmetic tasks: Previous studies have explored how models can benefit from incorporating precise numeric representations (Feng et al., 2024), or offloading the arithmetic computation to a tool (Schick et al., 2023; Kadlčík et al., 2023), suggesting that their native learned representations are not reliable. Other works (Kantamneni and Tegmark, 2025; Zhou et al., 2024) have inspected such learned representations directly and tried to understand how models use them. Although model probing methods showed some success in interpreting numeric values from model representations (Zhu et al., 2025), the accuracy of those methods is low, suggesting that learned representations are highly imprecise.

In this paper, we push back on this interpretation: we show that a probe with the *right kind of inductive bias* can retrieve numeric information from number embeddings with *near-perfect accuracy* across an extensive range of LMs, spanning the Llama 3 (Grattafiori et al., 2024), Phi 4 (Abdin et al., 2024) and OLMo 2 (OLMo et al., 2025) series and ranging from 1B to 72B parameters. Given that number embeddings usually follow a sinusoidal wave-like pattern (Nanda et al., 2023; Kantamneni and Tegmark, 2025), this characteristic must be accounted for when designing probes.

We further show how these insights can be leveraged to improve performances on arithmetic reasoning: errors on addition and subtraction tasks can often be matched with an inability of the probe to retrieve the expected numerical information for a given embedding, and demonstrate that intervening on number embeddings such that they more cohesively follow the pattern of other number embeddings can directly improve arithmetic performances. Lastly, we document edge cases that do not fall within this previously understood pattern: in particular, OLMo2 32B (OLMo et al., 2025) learns embeddings that are not sinusoidal-like, despite a high success rate on arithmetic tasks.

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2 Related Work

One line of work focuses on incorporating numerical values directly into token representations, providing LMs with a prior. Charton (2022) explores different number encodings based on scientific notation for training LM solvers of linear algebra problems. Golkar et al. (2023) propose representing numbers as a learned <NUM> token scaled by the number scalar value, demonstrating how models can adopt this scheme for regression tasks.

Another line of work investigates how models learn to represent and process numerical information. Nanda et al. (2023) show that a transformer with one-hot encoding trained from scratch on modular addition discovers Fourier basis and its computation is interpretable in trigonometric functions. Kantamneni and Tegmark (2025) discover an analogous circuitry for (non-modular) addition in a general pretrained language model, and find that its intermediate representations combine both linear and periodic components, reminiscent of a helix structure. Zhou et al. (2024) further identifies subcomponents of the addition circuitry implemented by the attention mechanism and feedforward layers. Zhu et al. (2025) demonstrate that hidden states of pretrained language models can be approximately decoded with a linear (or multi-layer) probe to estimate the logarithm of the number value. Although the probe outputs correlate with the target value, decoding achieves low accuracy. Recently, Levy and Geva (2025) show success in recovering the values of *digits* from internal representations of intermediate layers, hinting on a more generalized, circular pattern in representations of numbers.

In summary, prior works suggest that language models *attempt* to encode numerical information into token representations during pretraining, but their precision is rather limited. However, we hypothesize that this perception stems from inadequate probing methods, and learned representations are much more precise than previously estimated.

3 Recovering numerical information from number embeddings

We study LMs from the Llama 3 (Grattafiori et al., 2024), Phi 4 (Abdin et al., 2024), and OLMo 2 (OLMo et al., 2025) series, ranging from 1B to 72B parameters. Wide selection allows us to verify the validity of our observations across a panel of models sharing the characteristic of representing all integers between 0 and 999 with unique tokens.

Motivations. The central and foremost point to address is whether the embeddings representing specific numbers in LMs contain the numeric information of the value they represent. In practice, this is best addressed with a *probing* setup: If embeddings do contain numerical information, we should be able to learn a decoding function from number embedding to the corresponding integer value. Probing as a methodology comes with its own set of caveats: probes should be kept as simple as possible, and their expressivity should be compared against baseline benchmarks (Hewitt and Liang, 2019). Our specific use case adds further constraints: in particular, we have only one instance per LLM of each integer representation, viz., there is only one vector for the token `42`. This rules out naive classifier implementations, as we aim for the probe to generalize to entirely unseen classes.

Probe architectures. We consider four probes:

$$f_{\text{lin}}(\mathbf{x}) = \mathbf{a}^T\mathbf{x} + b \tag{1}$$

$$f_{\log\text{ lin}}(\mathbf{x}) = \exp\left(\mathbf{a}^T\mathbf{x} + b\right) - 1 \tag{2}$$

$$f_{\text{sin}}(\mathbf{x}) = (\mathbf{W}_{\text{out}}\mathbf{S})^T(\mathbf{W}_{\text{in}}\mathbf{x}) \tag{3}$$

$$f_{\text{bin}}(\mathbf{x}) = (\mathbf{W}_{\text{out}}\mathbf{B})^T(\mathbf{W}_{\text{in}}\mathbf{x}) \tag{4}$$

where $\mathbf{a}$, b, $\mathbf{W}_{\text{in}}$, and $\mathbf{W}_{\text{out}}$ are learned parameters, whereas $\mathbf{S}$ and $\mathbf{B}$ are means of injecting inductive biases in the linear classifiers f_{sin} and f_{bin}:

$$\mathbf{S}_{ij} = \begin{cases} \sin(ie^j 1000/d) & \text{if } j \equiv 0 \mod 2 \\ \cos(ie^{j+1} 1000/d) & \text{if } j \equiv 1 \mod 2 \end{cases}$$

$$\mathbf{B} = \begin{bmatrix} 0 & \dots & 0 & 0 & 1 \\ 0 & \dots & 0 & 1 & 0 \\ 0 & \dots & 0 & 1 & 1 \\ \vdots & & & & \end{bmatrix}$$

I.e., the ith row of $\mathbf{B}$ corresponds to the integer i expressed in binary, whereas $\mathbf{S}$ is defined as a Fourier basis, suggested by Zhou et al. as the hidden structure learned by pretrained models. The matrices $\mathbf{S}$ and $\mathbf{B}$ thus allows us to partition the label projection of the classifier into three components: a learned projection $\mathbf{W}_{\text{in}} : \mathbb{R}^d \to \mathbb{R}^h$ to project the number embeddings into a reduced low-dimensional space, a fixed matrix ($\mathbf{S}$ or $\mathbf{B}$) allowing us to encode integers using an *a priori* scheme, and a learned projection $\mathbf{W}_{\text{out}} : \mathbb{R}^d \to \mathbb{R}^h$ mapping these *a priori* representations onto the same space as the reduced embeddings. Intuitively, W_{in} uncovers the underlying hidden structure of the

learned embeddings, while W_{out} expresses it in terms of interpretable a priori basis, which allows us to generalize to unseen tokens.

Implementation. We evaluate the probes in Equations (1) to (4) using a cross-validation setup with 20 folds. We report their accuracy measured by rounding the output of the regression probes Equations (1) and (2) to the nearest integer, or by retrieving the index of the row in $\mathbf{S}$ or $\mathbf{B}$ that maximizes the output distribution of the classifier probes Equations (3) and (4). We control the validity of our probes by ensuring that they reach an accuracy of 0 for standard Gaussian vectors as well as for a random permutation of the embeddings. Parameters for regressions are estimated using a least-squares algorithm; whereas our classifiers' parameters are optimized with Adam with a learning rate of 0.0001, weight decay of 0.001, and $\beta = (0.9, 0.999)$. We choose a hidden dimension of 100. The classifiers are optimized to distinguish output *only between training tokens*, and during testing, must choose between all tokens. The probes are optimized until loss converges on a validation split separate from the testing split.

We release an implementation and training recipes for the new probes, including all configurations we use, in the project's GitHub repository.[1]

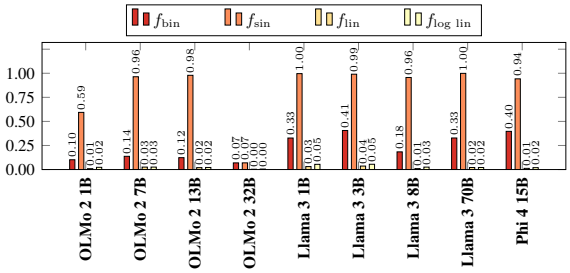

Figure 1: Overview of probes' accuracy (↑).

Results. We summarize performances, measured in terms of accuracy, in Figure 1. Crucially, we are almost systematically able to retrieve the integer value corresponding to the embedding's number with very high accuracy. Another salient observation is that f_{sin} consistently outperforms all other probe architectures including the regression probe used in previous work of Zhu et al. (2025), contradicting their finding that LMs learn to encode numbers linearly. Explaining the success of the Fourier basis, we note that other prior literature has suggested that sinusoidal features are used for arithmetic computation in LMs (Zhou et al., 2024). Adding onto this, we can also stress that, qualitatively, most of the models' whose number embeddings we survey here exhibit wave-like patterns in a PCA projection and have sparse Fourier transform, confirming regularity in the hidden structure. See Figures 2 and 3 in Section A.1 for visualizations of PCA and its Fourier transform. Notably, OLMo 2 32B is the only model with low resemblance of the pattern, which is consistent with the low performance of its sinusoidal probe.

Analysis. To verify that our sin-base probes indeed reach their superior accuracy by learning to extract a generalized, sin-like representation from models' representations, we analyse the encoded representations that trained sin probes produce as the output of $\mathbf{W}_{\text{in}}$. We experiment with two training settings: (i) using L1 regularization — encouraging sparsity, and (ii) using L2 regularization — encouraging the employment of a broader scale of input features. We note that in both of these settings, the probes achieve almost identical generalization capacity as assessed by their accuracy on unseen inputs (embeddings of numbers).

Figure 6 in Appendix A.3 displays the resulting representations for model embeddings of Llama3 1B associated with different numeric values. We can observe that the L1- and L2-regularized probes learn a substantially distinct representational pattern. We hypothesize that a main difference between probes trained with different regularizations is that the L1 probe learns to follow a broader scale of distinct frequencies, while the L2 probe follows similar frequencies shifted by a different constant. Nevertheless, in both of the cases, the probe learns a projection into a wave-like pattern across input numbers, thus successfully following their injected inductive bias.

4 Leveraging numerical information from number embeddings

Motivation. Having established that number embeddings do encode retrieval numerical information about the integers they represent, we now turn to **how this numerical information is leveraged** by LMs to perform arithmetic tasks. We study the zero-shot performances of a subset of our models on addition and subtraction tasks. We define our addition task as taking any pair of integers x_1, x_2 such that $0 < x_i < 500$ as input, and computing

[1] https://github.com/prompteus/numllama